Tanta University

Faculty of Computers and Information

Question Bank 2

The Whole Book in One Lecture: Artificial Intelligence

CS433 – Selected Topics in Artificial Intelligence

Complete Solved Bank
30 Multiple-Choice Questions + 10 True/False Questions
Original English questions and options, correct answers, English explanations, Arabic explanations, and lecture references.

Prepared for study and revision

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1 Part A – Multiple Choice

Choose the one best answer.

A1. Which of the four approaches to AI does the textbook adopt?

(a) Thinking humanly

(b) Acting rationally

(c) Acting humanly

(d) Thinking rationally

Correct Answer: (b) Acting rationally.

Explanation (English): The textbook adopts the rational-agent approach. It defines AI around agents that choose actions to maximise the expected value of their performance measure.

الشرح بالعربي: الكتاب بيعتمد مدخل Acting، Rationally يعني بنركز على Agent يتصرف بطريقة عقلانية ويختار الفعل اللي يحقق أعلى أداء متوقع بناءً على المعلومات المتاحة له.

Reference: CS433-W00-Lecture-Notes, Section 0.1 – Four approaches to AI and the standard model.

A2. The Turing test (1950) belongs to which approach to AI?

(a) Thinking rationally

(b) Thinking humanly

(c) Acting rationally

(d) Acting humanly

Correct Answer: (d) Acting humanly.

Explanation (English): The Turing test evaluates whether a machine can behave in conversation in a way that is indistinguishable from a human. It is therefore part of the acting-humanly approach.

الشرح بالعربي: اختبار Turing بيقيس هل الآلة تقدر تتصرف في المحادثة بطريقة تخلي الشخص اللي بيختبرها مش قادر يميزها عن الإنسان؛ عشان كده هو Acting، Humanly.

Reference: CS433-W00-Lecture-Notes, Section 0.1.

A3. Under the standard model, a rational agent chooses the action that:

(a) maximises the expected value of its performance measure, given what it has perceived so far

(b) is guaranteed to give the best actual outcome

(c) most closely copies what a human would do

(d) follows from Aristotle's syllogisms

Correct Answer: (a) maximises the expected value of its performance measure, given what it has perceived so far.

Explanation (English): The standard model defines rational action as maximising expected performance based on the percept history available to the agent. Rationality does not require knowing or guaranteeing the actual future outcome.

الشرح بالعربي: الـ Agent Rational يختار الفعل اللي يحقق أكبر قيمة متوقعة لمقياس الأداء، اعتمادًا على كل اللي أدركه لحد اللحظة دي. مش شرط يضمن أفضل نتيجة فعلية؛ لأنه ممكن يكون فيه عدم يقين.

Reference: CS433-W00-Lecture-Notes, Section 0.1 – Standard model.

A4. The danger that a machine pursuing a fixed objective does harm because the objective is slightly wrong is called:

(a) the combinatorial explosion

(b) the curse of dimensionality

(c) the value alignment problem

(d) the frame problem

Correct Answer: (c) the value alignment problem.

Explanation (English): The value alignment problem concerns the risk that a machine optimising a specified objective may cause harm if that objective does not accurately reflect human values or intentions.

الشرح بالعربي: مشكلة Alignment Value معناها إن الآلة ممكن تنفذ الهدف المطلوب منها بكفاءة، لكن لو الهدف نفسه مش معبّر بدقة عن قيم الإنسان أو قصده، ممكن النتيجة تكون مؤذية رغم إن الآلة نفذت التعليمات.

Reference: CS433-W00-Lecture-Notes, Section 0.1 – Limited rationality and value alignment.

A5. In which year did the Dartmouth workshop give the field of AI its name?

(a) 1943

(b) 1956

(c) 1969

(d) 1986

Correct Answer: (b) 1956.

Explanation (English): The Dartmouth workshop took place in 1956 and is identified in the historical timeline as the event that named the field of artificial intelligence.

الشرح بالعربي: ورشة Dartmouth كانت سنة 1956، ودي السنة المرتبطة بتسمية مجال Artfiicial Intelligence.

Reference: CS433-W00-Lecture-Notes, Section 0.1 – Foundations and history.

A6. According to the notes, an agent is made of:

(a) architecture + program

(b) sensors + critic

(c) percepts + goals

(d) knowledge base + utility

Correct Answer: (a) architecture + program.

Explanation (English): An agent program implements the agent function and runs on a physical architecture. The notes summarise this as agent = architecture + program.

الشرح بالعربي: الـ Agent بيتكوّن من Architecture، يعني الجهاز أو البيئة المادية اللي بيشتغل عليها، وProgram اللي بينفذ سلوك الـ Agent.

Reference: CS433-W00-Lecture-Notes, Section 0.2 – Intelligent agents.

A7. Rationality depends on four things. Which of the following is NOT one of them?

(a) The performance measure

(b) The agent's prior knowledge

(c) The percept sequence so far

(d) Knowing the actual outcome of every action in advance

Correct Answer: (d) Knowing the actual outcome of every action in advance.

Explanation (English): Rationality depends on the performance measure, prior knowledge, available actions, and the percept sequence so far. It does not require foreknowledge of the actual outcome of every action.

الشرح بالعربي: العقلانية بتعتمد على مقياس الأداء، والمعرفة السابقة، والأفعال المتاحة، وتسلسل الإدراكات لحد دلوقتي. مش مطلوب من الـ Agent يعرف المستقبل أو النتيجة الفعلية لكل فعل قبل ما ينفذه.

Reference: CS433-W00-Lecture-Notes, Section 0.2 – Rationality.

A8. In the PEAS description of the automated taxi, "steering, accelerator, brake, signal, horn" are the:

(a) Performance measure

(b) Environment

(c) Actuators

(d) Sensors

Correct Answer: (c) Actuators.

Explanation (English): Actuators are the mechanisms through which an agent acts on its environment. Steering, accelerator, brake, signal, and horn are outputs used by the taxi to act.

الشرح بالعربي: الـ Actuators هي وسائل تنفيذ الأفعال والتأثير في البيئة. الدركسيون والبنزين والفرامل والإشارة والكلاكس كلها أدوات التاكسي اللي بيستخدمها عشان يتحرك ويتصرف.

Reference: CS433-W00-Lecture-Notes, Section 0.2 – PEAS and automated taxi.

A9. In the table of environment dimensions, poker is given as an example of an environment that is:

(a) partially observable

(b) deterministic

(c) single-agent

(d) static

Correct Answer: (a) partially observable.

Explanation (English): Poker is partially observable because each player cannot see the other players' private cards. The agent must make decisions with incomplete information.

الشرح بالعربي: Poker بيئة Observable Partially لأن كل لاعب مش شايف الورق اللي مع باقي اللاعبين. يعني القرار بيتاخد بمعلومات ناقصة، مش كل حالة اللعبة ظاهرة له.

Reference: CS433-W00-Lecture-Notes, Section 0.2 – Environment dimensions.

A10. Which agent program fails when the right action depends on history?

(a) Utility-based agent

(b) Simple reflex agent

(c) Goal-based agent

(d) Model-based reflex agent

Correct Answer: (b) Simple reflex agent.

Explanation (English): A simple reflex agent selects actions using only the current percept and condition–action rules. It has no internal state for remembering relevant history, so it can fail when past percepts matter.

الشرح بالعربي: الـ Agent Reflex Simple بياخد قراره من الـ Percept Current بس، ومش بيحتفظ بتاريخ اللي حصل قبل كده. فلو الفعل الصح محتاج معلومات من الماضي، النوع ده ممكن يفشل.

Reference: CS433-W00-Lecture-Notes, Section 0.2 – Agent programs.

A11. In a learning agent, which component gives feedback on how well the agent is doing?

(a) The performance element

(b) The problem generator

(c) The actuator

(d) The critic

Correct Answer: (d) The critic.

Explanation (English): The critic evaluates the agent's behaviour against a performance standard and provides feedback to the learning element.

الشرح بالعربي: الـ Critic وظيفته يقيّم أداء الـ Agent ويقول له هو ماشي كويس ولا محتاج يتحسن، وبعدها الـ Element Learning يستفيد من الـ Feedback ده.

Reference: CS433-W00-Lecture-Notes, Section 0.2 – Learning agents.

A12. In a factored representation, a state is:

(a) an indivisible label

(b) a set of objects and the relations among them

(c) a vector of variable values

(d) a single number giving its utility

Correct Answer: (c) a vector of variable values.

Explanation (English): A factored representation describes a state using a collection of variables, with each state represented by the values assigned to those variables.

الشرح بالعربي: في الـ Representation Factored بنمثل الحالة بمجموعة Variables، وكل حالة بتكون عبارة عن قيم المتغيرات دي، يعني ممكن نكتبها كـ Vector من القيم.

Reference: CS433-W00-Lecture-Notes, Section 0.3 – State representations.

A13. Which of the following is NOT one of the five parts of a search problem?

(a) The initial state

(b) A heuristic function

(c) The transition model Result(s, a)

(d) The goal test

Correct Answer: (b) A heuristic function.

Explanation (English): The five components of a search problem are the initial state, actions, transition model, goal test, and path-cost function. A heuristic function can guide informed search, but it is not one of the five basic components.

الشرح بالعربي: مكونات Problem Search الأساسية خمسة: Initial State، Actions Transition، Model Goal، Test Path، وCost. الـ Function Heuristic بتساعد في البحث الموجّه زي A*، لكنها مش واحدة من الخمسة الأساسية.

Reference: CS433-W00-Lecture-Notes, Section 0.3 – Problem formulation.

A14. What is the space complexity of depth-first search, with branching factor b and maximum depth m?

(a) $O(bm)$

(b) $O(b^m)$

(c) $O(b^d)$

(d) $O(bd)$

Correct Answer: (a) $O(bm)$.

Explanation (English): Depth-first search stores the current path and the unexplored siblings along that path. Its standard space complexity is $O(bm)$, where b is the branching factor and m is the maximum depth.

الشرح بالعربي: في DFS مش بنخزن كل الشجرة، بنخزن المسار الحالي وبعض العقد الشقيقة اللي لسه هنزورها. عشان كده المساحة القياسية هي $O(bm)$، حيث b عدد الفروع وm أقصى عمق.

Reference: CS433-W00-Lecture-Notes, Section 0.3 – Search algorithm complexity.

A15. In the Romania A* example, Sibiu has $g = 140$ and $h = 253$. What is its f value?

(a) 113

(b) 253

(c) 393

(d) 447

Correct Answer: (c) 393.

Explanation (English): A* evaluates a node using $f(n) = g(n) + h(n)$. Therefore, $f(\text{Sibiu}) = 140 + 253 = 393$.

الشرح بالعربي: في خوارزمية A* بنحسب $f = g + h$، يعني تكلفة الوصول الفعلية لحد العقدة زائد التقدير المتبقي للهدف. هنا $140 + 253 = 393$.

Reference: CS433-W00-Lecture-Notes, Section 0.3 – A on the Romania map.*

A16. How long is the optimal route from Arad to Bucharest that A* finds?

(a) 393 km

(b) 450 km

(c) 449 km

(d) 418 km

Correct Answer: (d) 418 km.

Explanation (English): The optimal route found by A* is Arad–Sibiu–Rimnicu Vilcea–Pitesti–Bucharest, with a total distance of 418 km. The 450 km route is the one found by greedy best-first search in the notes.

الشرح بالعربي: المسار الأمثل هو Arad ثم Sibiu ثم Vilcea Rimnicu ثم Pitesti ثم Bucharest، وإجمالي المسافة 418 كم. خلي بالك إن 450 كم هو مسار Best-First Greedy مش المسار الأمثل اللي A* بيجيبه.

Reference: CS433-W00-Lecture-Notes, Section 0.3 – Romania A worked example.*

A17. A heuristic $h(n)$ is admissible if it:

(a) is always equal to zero

(b) never overestimates the true cost to reach a goal

(c) always overestimates the true cost to reach a goal

(d) is learned from data

Correct Answer: (b) never overestimates the true cost to reach a goal.

Explanation (English): An admissible heuristic is optimistic: its estimated cost to the goal is never greater than the true minimum remaining cost. This property supports A* optimality under the relevant search conditions.

الشرح بالعربي: الـ Heuristic بتكون Admissible لو عمرها ما تبالغ في تقدير التكلفة الحقيقية المتبقية للوصول للهدف، يعني التقدير يكون أقل من أو يساوي التكلفة الفعلية.

Reference: CS433-W00-Lecture-Notes, Section 0.3 – A and admissible heuristics.*

A18. When actions are nondeterministic, Chapter 4 uses AND–OR search. What form does a solution take?

(a) A contingency plan (if–then)

(b) A single fixed sequence of actions

(c) A set of belief states

(d) A gradient vector

Correct Answer: (a) A contingency plan (if–then).

Explanation (English): With nondeterministic actions, one action can lead to several possible outcomes. An AND–OR solution therefore specifies what to do for each possible outcome, forming a contingency plan.

الشرح بالعربي: لما الفعل يكون Nondeterministic ممكن ينتج عنه أكتر من نتيجة. عشان كده الحل مش سلسلة أفعال ثابتة، بنعمل خطة شرطية: لو حصلت النتيجة الأولى نعمل كذا، ولو حصلت التانية نعمل كذا.

Reference: CS433-W00-Lecture-Notes, Section 0.4 – AND–OR search.

A19. In backtracking search for CSPs, the minimum-remaining-values heuristic:

(a) tries the friendliest value first

(b) breaks ties by the most constraints

(c) chooses the most constrained variable

(d) chooses variables in random order

Correct Answer: (c) chooses the most constrained variable.

Explanation (English): Minimum Remaining Values (MRV) selects the unassigned variable with the fewest legal values remaining. This is the most constrained variable and helps expose failure early.

الشرح بالعربي: MRV بتختار المتغير اللي فاضل له أقل عدد من القيم المسموح بيها. يعني بنبدأ بالمتغير الأكثر تقييدًا عشان لو فيه تعارض نكتشفه بدري.

Reference: CS433-W00-Lecture-Notes, Section 0.5 – CSP heuristics.

A20. In the minimax example, the left MIN node sees 3 and 12, and the right MIN node's first child is 2. What is the value of the root?

(a) 2

(b) 12

(c) 0

(d) 3

Correct Answer: (d) 3.

Explanation (English): The left MIN node returns $\min(3, 12) = 3$. The right MIN node has a child with value 2, so its value cannot exceed 2. The root is a MAX node and therefore selects $\max(3, \text{right MIN value}) = 3$.

الشرح بالعربي: عقدة MIN اللي على الشمال بتختار الأصغر بين 3 و12، يعني قيمتها 3. عقدة MIN اللي على اليمين عندها ابن قيمته 2، فمش هتقدر ترجع قيمة أكبر من 2. الجذر MAX هيختار الأكبر بين 3 وقيمة اليمين، وبالتالي الناتج 3.

Reference: CS433-W00-Lecture-Notes, Section 0.6 – Minimax.

A21. With perfect move ordering, α-β pruning examines about how many nodes?

(a) $O(b^{m/2})$

(b) $O(b^{m})$

(c) $O(bm)$

(d) $O(2m)$

Correct Answer: (a) $O(b^{m/2})$.

Explanation (English): With ideal move ordering, alpha–beta pruning reduces the effective search depth roughly by half. Its best-case number of examined nodes is commonly expressed as $O(b^{m/2})$.

الشرح بالعربي: لما ترتيب الحركات يكون مثالي، Pruning Alpha-Beta بتقلل عدد العقد اللي محتاجين نفحصها بشكل كبير، والتعقيد الأفضل بيكون تقريبًا $O(b^{m/2})$ بدل $O(b^m)$.

Reference: CS433-W00-Lecture-Notes, Section 0.6 – Minimax and alpha-beta pruning.

A22. $KB \models \alpha$ holds exactly when:

(a) $KB \wedge \alpha$ is unsatisfiable

(b) $KB \wedge \neg\alpha$ is unsatisfiable

(c) α is true in at least one model of KB

(d) $KB \Rightarrow \alpha$ is unsatisfiable

Correct Answer: (b) $KB \wedge \neg\alpha$ is unsatisfiable.

Explanation (English): Knowledge base KB entails α if every model satisfying KB also satisfies α. Equivalently, there is no model in which KB is true and α is false; hence $KB \wedge \neg\alpha$ is unsatisfiable.

الشرح بالعربي: معنى إن KB بتستلزم Alpha هو إن كل Model يحقق الـ KB لازم يحقق Alpha كمان. بطريقة تانية: مستحيل نلاقي حالة يكون فيها KB صح وAlpha غلط، لذلك $KB \wedge \neg\alpha$ لازم تكون Unsatisfiable.

Reference: CS433-W00-Lecture-Notes, Section 0.7 – Logical agents and entailment.

A23. What is Unify(Knows(John, x), Knows(y, Bill))?

(a) $\{x/John,\ y/Bill\}$

(b) $\{x/y\}$

(c) $\{x/Bill,\ y/John\}$

(d) No unifier exists

Correct Answer: (c) $\{x/Bill,\ y/John\}$.

Explanation (English): To unify the two expressions, match corresponding arguments. The first arguments require $y = John$, and the second require $x = Bill$. Thus the substitution is $\{y/John,\ x/Bill\}$, which corresponds to option (c), not option (a).

الشرح بالعربي: هنا لازم نطابق الوسيط الأول مع الأول والتاني مع التاني: في المكان الأول John لازم تساوي y، وفي المكان التاني x لازم تساوي Bill. إذن الـ substitution الصحيح هو $\{x/Bill,\ y/John\}$، يعني الاختيار (c).

Reference: CS433-W00-Lecture-Notes, Section 0.8 – First-order logic and unification.

A24. In the meningitis example, $P(s \mid m) = 0.7$, $P(m) = 1/50,000$ and $P(s) = 0.01$. What is $P(m \mid s)$?

(a) 0.7

(b) 0.07

(c) 0.014

(d) 0.0014

Correct Answer: (d) 0.0014.

Explanation (English): By Bayes' rule,

$$P(m \mid s) = \frac{P(s \mid m)P(m)}{P(s)} = \frac{0.7 \times (1/50{,}000)}{0.01} = \frac{0.000014}{0.01} = 0.0014.$$

الشرح بالعربي: نطبق قانون Bayes: احتمال المرض مع وجود العرض = احتمال العرض مع المرض × الاحتمال الأساسي للمرض، الكل على احتمال العرض. بالتعويض: $0.7 \times (1/50000) \div 0.01 = 0.0014$.

Reference: CS433-W00-Lecture-Notes, Section 0.11 – Bayes' rule, meningitis worked example.

A25. How many numbers does the burglary Bayesian network (five Boolean variables) need, compared with the full joint table?

(a) 5 versus 32

(b) 10 versus 31

(c) 31 versus 10

(d) 10 versus 25

Correct Answer: (b) 10 versus 31.

Explanation (English): A full joint distribution over five Boolean variables has $2^5 = 32$ entries, but one is determined by the requirement that probabilities sum to 1, leaving 31 independent numbers. In the burglary network, the conditional probability tables require 10 numbers, exploiting conditional independence.

الشرح بالعربي: الـ Table Joint Full لخمس Variables كل واحدة Boolean فيها $2^5 = 32$ احتمال، لكن واحد منهم بيتحدد من شرط إن المجموع يساوي واحد، فيتبقى 31 رقم مستقل. شبكة Burglary بتحتاج 10 أرقام بس بسبب استغلال Conditional Independence.

Reference: CS433-W00-Lecture-Notes, Section 0.12 – Bayesian networks and compact representations.

A26. Which algorithm computes the most likely explanation $\arg\max_{x_{1:t}} P(x_{1:t} \mid e_{1:t})$ in a temporal model?

(a) Viterbi

(b) The forward algorithm

(c) Forward–backward

(d) Value iteration

Correct Answer: (a) Viterbi.

Explanation (English): The Viterbi algorithm finds the most probable sequence of hidden states given the observations. The forward algorithm computes filtering probabilities, while forward–backward computes smoothed state probabilities.

الشرح بالعربي: Viterbi بتدور على أكتر تسلسل States Hidden احتمالًا في النموذج الزمني بناءً على الـ Evidence. أما Forward فبتحسب احتمالات الحالة الحالية، وForward-Backward بتستخدم المعلومات السابقة واللاحقة لعمل Smoothing.

Reference: CS433-W00-Lecture-Notes, Section 0.13 – Temporal models and hidden Markov models.

A27. In the prisoner's dilemma, both prisoners testifying is:

(a) Pareto optimal but not a Nash equilibrium

(b) neither a Nash equilibrium nor a dominant strategy

(c) worse for Ali than refusing, whatever Bo does

(d) a Nash equilibrium but not Pareto optimal

Correct Answer: (d) a Nash equilibrium but not Pareto optimal.

Explanation (English): Testifying is a dominant strategy for each prisoner, so both testifying is a Nash equilibrium: neither can improve by changing strategy alone. However, both refusing gives each a better outcome, so the equilibrium is not Pareto optimal.

الشرح بالعربي: لما الاتنين يشهدوا، دي Equilibrium Nash لأن مفيش واحد فيهم يقدر يحسّن نتيجته لو غيّر قراره لوحده. لكن النتيجة مش Pareto Optimal؛ لأن لو الاتنين رفضوا الشهادة، الاتنين هيكونوا في وضع أفضل.

Reference: CS433-W00-Lecture-Notes, Section 0.14 – Game theory and prisoner's dilemma.

A28. In the restaurant problem, what is the information gain of splitting on Patrons?

(a) 0 bits

(b) 1 bit

(c) about 0.541 bits

(d) about 0.918 bits

Correct Answer: (c) about 0.541 bits.

Explanation (English): Information gain is the entropy before the split minus the weighted average entropy after the split. For the restaurant example in the course material, splitting on Patrons gives approximately 0.541 bits.

الشرح بالعربي: Gain Information معناها مقدار الانخفاض في الـ Entropy بعد ما نقسم البيانات باستخدام Attribute معين. في مثال المطعم الموجود في المنهج، التقسيم باستخدام Patrons يحقق تقريبًا 0.541 bits.

Reference: CS433-W00-Lecture-Notes, Section 0.16 – Learning from examples and information gain.

A29. One Q-learning update uses $Q(s,a) = 0.5$, $R = -0.04$, $\max_{a'} Q(s',a') = 0.8$, $\gamma = 1$ and $\alpha = 0.1$. What is the new $Q(s,a)$?

(a) 0.500

(b) 0.526

(c) 0.760

(d) 0.804

Correct Answer: (b) 0.526.

Explanation (English): Use the Q-learning update:

$$Q_{\text{new}}(s,a) = Q(s,a) + \alpha\left[R + \gamma \max_{a'} Q(s',a') - Q(s,a)\right].$$

Substitute the values:

$$= 0.5 + 0.1[-0.04 + (1)(0.8) - 0.5] = 0.5 + 0.1(0.26) = 0.526.$$

الشرح بالعربي: قانون التحديث: القيمة الجديدة = القديمة + Rate Learning × (المكافأة + Gamma × أفضل Q في الحالة التالية − القيمة القديمة). بالتعويض: $0.5 + 0.1(-0.04 + 0.8 - 0.5) = 0.526$.

Reference: CS433-W00-Lecture-Notes, Section 0.19 – Reinforcement learning and Q-learning.

A30. In robotics (Chapter 26), localisation and mapping (SLAM) is done with which tools?

(a) Kalman and particle filters

(b) The CYK chart parser

(c) Minimax and α-β pruning

(d) PDDL action schemas

Correct Answer: (a) Kalman and particle filters.

Explanation (English): SLAM combines localisation (estimating the robot's position) and mapping (building a map of the environment). Probabilistic state-estimation tools such as Kalman filters and particle filters are used for these tasks.

الشرح بالعربي: اختصار SLAM يعني Simultaneous Localisation and Mapping، يحدد الروبوت مكانه وفي نفس الوقت يبني خريطة للمكان. من الأدوات الاحتمالية المستخدمة في تقدير الحالة Kalman Filters وParticle Filters.

Reference: CS433-W00-Lecture-Notes, Part VI – Perception and robotics (Chapter 26).

2 Part B – True or False

State whether each statement is true or false.

B1. Because perfect rationality is usually computationally impossible, the book aims for limited rationality.

Correct Answer: True.

Explanation (English): The notes state that perfect rationality is usually computationally impossible, motivating the aim of limited rationality.

الشرح بالعربي: العبارة صح. الوصول لعقلانية مثالية في كل موقف غالبًا مستحيل حسابيًا، عشان كده الكتاب بيتكلم عن Limited Rationality، أي عقلانية محدودة بالموارد والوقت المتاحين.

Reference: CS433-W00-Lecture-Notes, Section 0.1 – Limited rationality.

B2. In the Romania example, greedy best-first search finds the same 418 km route as A*.

Correct Answer: False.

Correct statement: Greedy best-first search finds the 450 km route, whereas A* finds the optimal 418 km route.

Explanation (English): The notes explicitly contrast the two results: A* finds the 418 km optimal route, while greedy best-first search takes the 450 km route through Fagaras.

الشرح بالعربي: العبارة غلط. A* هي اللي بتوصل للمسار الأمثل 418 كم، لكن Best-First Greedy بتختار مسار 450 كم عن طريق Fagaras.

Reference: CS433-W00-Lecture-Notes, Section 0.3 – Romania map example.

B3. Local search with min-conflicts solves the million-queens problem in about 50 steps.

Correct Answer: True.

Explanation (English): The course notes state that local search using the min-conflicts heuristic solves the million-queens problem in about 50 steps.

الشرح بالعربي: العبارة صح حسب المحاضرة؛ خوارزمية Min-Conflicts في Local Search قادرة على حل مشكلة مليون ملكة في حوالي 50 خطوة.

Reference: CS433-W00-Lecture-Notes, Section 0.5 – CSPs and min-conflicts.

B4. Entailment in first-order logic is decidable: a procedure can always confirm that a sentence is not entailed.

Correct Answer: False.

Correct statement: Entailment in first-order logic is semi-decidable: if a sentence is entailed, a procedure can eventually confirm it, but it may not halt when the sentence is not entailed.

Explanation (English): First-order logic validity and entailment are semi-decidable, not fully decidable. A proof procedure can find a proof when one exists, but failure to find a proof does not necessarily establish non-entailment.

الشرح بالعربي: العبارة غلط. الـ Entailment في Logic First-Order مش Decidable بالكامل؛ هو Semi-decidable. يعني لو الجملة مستلزمة فعلًا ممكن إجراء البرهان يوصل لإثباتها، لكن لو مش مستلزمة ممكن البحث يفضل شغال من غير ما ينتهي، ومش دايمًا نقدر نأكد النفي.

Reference: CS433-W00-Lecture-Notes, Sections 0.7–0.8 – Logic and first-order logic.

B5. The naive Bayes model assumes all effects are independent given one cause, so it grows only linearly.

Correct Answer: True.

Explanation (English): Naive Bayes assumes conditional independence of the effects given the class or common cause. This factorisation greatly reduces the number of parameters, making the model grow linearly with the number of effects.

الشرح بالعربي: العبارة صح. Naive Bayes بتفترض إن الـ Effects مستقلة عن بعضها بشرط معرفة الـ Cause أو Class. الافتراض ده بيقلل عدد الاحتمالات اللي محتاجين نخزنها، فالنمو بيكون خطي بدل ما يكبر بشكل أُسّي.

Reference: CS433-W00-Lecture-Notes, Section 0.11 – Conditional independence and naive Bayes.

B6. If a machine is unsure what the human wants, it has a positive incentive to let the human switch it off.

Correct Answer: True.

Explanation (English): Under the uncertain-objective framework described in the notes, a machine that is uncertain about human preferences has an incentive to preserve human control, including allowing itself to be switched off.

الشرح بالعربي: العبارة صح. لو الآلة مش متأكدة الإنسان عايز إيه بالضبط، فمن مصلحتها إنها تسيب للإنسان القدرة على إيقافها، لأنها ممكن تكون فاهمة الهدف غلط، فالأفضل تحافظ على تحكم الإنسان.

Reference: CS433-W00-Lecture-Notes, Section 0.14 – Uncertain preferences and safety.

B7. In the book's 4×3 grid world, the agent moves as intended with probability 0.9 and slips sideways with probability 0.05 each way.

Correct Answer: False.

Correct statement: In the course notes' 4×3 grid world, the agent moves as intended with probability 0.8 and slips sideways with probability 0.1 each way.

Explanation (English): The lecture notes specify an intended-move probability of 0.8 and sideways-slip probabilities of 0.1 in each direction. The values 0.9 and 0.05 are not the values stated in the course notes.

الشرح بالعربي: العبارة غلط حسب أرقام المحاضرة: الحركة في الاتجاه المقصود احتمالها 0.8، والانزلاق لكل اتجاه جانبي احتماله 0.1، مش 0.9 و0.05.

Reference: CS433-W00-Lecture-Notes, Section 0.14 – Markov decision processes, 4x3 grid world.

B8. Maximum-likelihood learning averages all hypotheses weighted by their posterior probability.

Correct Answer: False.

Correct statement: Bayesian learning averages hypotheses weighted by their posterior probabilities; maximum-likelihood learning selects the hypothesis that best explains the observed data.

Explanation (English): The statement describes Bayesian learning, not maximum likelihood. Maximum likelihood chooses the hypothesis with the highest likelihood of the observed data, whereas Bayesian learning averages predictions over hypotheses weighted by their posterior probabilities.

الشرح بالعربي: العبارة غلط لأنها خلطت بين طريقتين. Bayesian Learning بتاخد متوسط الفرضيات مع وزن كل فرضية حسب Posterior Probability. Maximum Likelihood إنما بتختار الفرضية اللي بتفسر البيانات المرصودة بأعلى احتمال.

Reference: CS433-W00-Lecture-Notes, Section 0.17 – Learning as probabilistic inference.

B9. Searle's Chinese room argues that symbol manipulation is not the same as understanding.

Correct Answer: True.

Explanation (English): The Chinese Room argument distinguishes syntactic manipulation of symbols from semantic understanding. Following rules to manipulate symbols does not, by itself, establish genuine understanding.

الشرح بالعربي: العبارة صح. حجة الغرفة الصينية لـ Searle بتقول إن مجرد التعامل مع الرموز واتباع قواعد لمعالجتها مش معناه بالضرورة إن النظام فاهم المعنى فعلًا.

Reference: CS433-W00-Lecture-Notes, Part VII – AI philosophy and ethics.

B10. Weak AI is the claim that machines would actually have minds.

Correct Answer: False.

Correct statement: Strong AI is the claim that machines could actually have minds; Weak AI treats machines as tools that can behave intelligently without claiming that they literally have minds.

Explanation (English): The statement describes the strong-AI position. Weak AI claims that machines can simulate or exhibit intelligent behaviour, without asserting that they possess genuine minds or consciousness.

الشرح بالعربي: العبارة غلط. فكرة إن الآلات ممكن يكون عندها عقول فعلية دي Strong AI. أما Weak AI فبتتعامل مع الآلات كأنها أدوات تقدر تنفذ سلوك ذكي، من غير ما نزعم إنها تمتلك عقل أو وعي حقيقي.

Reference: CS433-W00-Lecture-Notes, Part VII – AI philosophy and ethics.

3 Quick Answer Key

Part A – Multiple Choice

Question	Answer	Question	Answer	Question	Answer	Question	Answer
A1	B	A2	D	A3	A	A4	C
A5	B	A6	A	A7	D	A8	C
A9	A	A10	B	A11	D	A12	C
A13	B	A14	A	A15	C	A16	D
A17	B	A18	A	A19	C	A20	D
A21	A	A22	B	A23	C	A24	D
A25	B	A26	A	A27	D	A28	C
A29	B	A30	A				

Part B – True or False

Question	Answer	Question	Answer	Question	Answer	Question	Answer
B1	True	B2	False	B3	True	B4	False
B5	True	B6	True	B7	False	B8	False
B9	True	B10	False				

End of Question Bank 2 – Week 0